BC Salmon Restoration and Innovation Fund (BCSRIF)
Year-End Report Form

To be submitted with the Year-End (March 31) Claim for FY 2022-2023

The BC Salmon Restoration and Innovation Fund (BCSRIF) requires funding recipients to complete this form as part of their year-end reporting requirements. Projects funded through BCSRIF, which are undertaken by recipients, form the program's achievements; in short, the recipient narrative supports the success of BCSRIF.

If this is your FINAL project year, please capture all of the excellent benefits, results and outcomes of your project, such that BCSRIF can showcase that information in the Annual Results Summary Report for the fiscal year and BCSRIF's overall program achievements can be captured for Departmental reporting to the Canadian public.

The data collected through year-end reporting will complement other information provided previously by recipients through application documents and the development of the Contribution Agreement. **The information provided is not intended to replace any project-specific progress or final reporting outlined in the Contribution Agreement.**

Results obtained via this form will be used to support BCSRIF in reporting annually to its government partners, funding recipients, and the public on individual project achievements and overall program outcomes. The data will be used to support internal analysis of program success and address potential gaps in the suite of funding programs delivered across the Pacific Region. **Annual reports will be produced by BCSRIF to highlight key project achievements and provide results on how BCSRIF projects are benefiting BC's fish and seafood sector and contributing to the sustainability of wild Pacific salmon and other BC fish stocks.**

The results of this report form are depersonalized. Please refrain from including identifiable information such as personal information in your response. Information received throughout this submission process is subject to the *Access to Information Act* and the *Privacy Act*.

Note that general information about approved projects, including the names of recipients, the purpose of the funding, and total funding approved, are included in public records and published in accordance with the Government of Canada's proactive disclosure requirements.

Completed forms should be submitted with your year-end claim for payment. If you have any questions regarding this form or how the information will be used, please contact your Program Officer, or contact the BCSRIF team at DFO.BCSRIF-FRISCB.MPO@dfo-mpo.gc.ca.

Section 1: Project Information

This section is intended to capture data that describes your project's scope, scale, and location information.

1. **Scale of Project Influence [Choose one only]**
   ☐Community (local area streams)
   ☐Regional (major stream systems/ multi-watershed)
   ☒Provincial (BC-wide applicability)
   ☐National (Canada-wide applicability)
   ☐International

2. **Detailed geographic information on project site location(s), list all as applicable.**

   This section is meant to provide a full list of all projects and sub-projects completed to date, until project completion. Please include all of your projects and sub-projects to date.

   BCSRIF will use project GPS coordinate information to map and highlight funded projects on our website. BCSRIF also requests you provide representative photo(s) of your project that can be used for a pop-up box summary description, if available. Please attach to your submission.

Project or Site Name	**Name of River or Watershed**	**Nearest community**	**GPS coordinates** [decimal degrees, preferred format]	**Year completed**
BC EEZ				
Waters around Galiano Island		Galiano Island		
Strait of Georgia		Nanaimo		
[add rows, as needed]				

Section 2: Demographic Information

Information on demographic representation enables the Government of Canada to examine the influence of government programs on diversity and inclusion. Data collected on the various intersecting identify factors will help provide information in support of Gender-Based Analysis Plus (GBA+) associated with Grant and Contribution programs. We request you indicate if persons on your project team represent any of the following demographic groups, where known. Data collected through this

section will be summarized, and reported at an aggregate level. As stated above, please refrain from including details that link responses to an individual.

3. **Primary project team includes members of the following demographic groups [select all that apply]:**

☐Women

☐Indigenous peoples

☐LGBTQ2+ community

☒Ethnic, linguistic, cultural or religious minority

☐Seniors 65+

☐Veterans

☒Students or youth or children

☐Persons with disabilities

☐New Canadians

4. **Does your project include components that specifically target/encourage participation by representatives of any of the demographic groups listed above? [Please provide a short description]:** *For example, a key element of an early BCSRIF project aims to increase participation by youth and new Canadians in the recreational fishing sector, through outreach and engagement on employment and other economic development opportunities.*

Not specifically. We are targeting all recreational fishers in BC's tidal waters.

Section 3: Socio-economic Project Metrics

Data related to the project metrics in this section are collected to enable BCSRIF to report out on business and partnership-related metrics in support of this program and to inform the potential development of future Grants and Contributions funding programs. These questions are consistent with annual recipient reporting requirements for other regional grant and contribution programs. Where exact numbers are unavailable, we ask that you estimate answers to the best of your knowledge.

5. **Socio-economic Performance Measures**

Performance Measure	Total Number during Fiscal Year of reporting	Cumulative Total for project to date
Partnerships		

Number of total partner organizations engaged in the project (i.e., providing cash and/ or in-kind support to project) – *do not count BCSRIF program staff or CA recipient in total*	5	5
Number of Indigenous partner organizations engaged in the project (i.e., providing cash and/ or in-kind support to project)	0	0
Number of total volunteer persons engaged in the project		
Employment		
Total number of persons paid through BCSRIF funding (i.e., employed directly full time, part time or by direct-award contract)	7	31
Number of Indigenous persons employed by the project (directly or indirectly, full time, part time or by contract)	1	1
Average salary of persons employed directly by the project	NA	
Number of full-time employees (persons employed > or = 30 hrs/ week)	0	0
Number of persons in long-term employment positions (employed > or = 6 months)	4	17
Number of persons employed in rural or remote communities (community with population < 1000 inhabitants)	0	0
Training		
Number of people trained in association with delivering the project (in any area of skills development/ expertise to enable the project to proceed including fisheries or habitat assessment, monitoring, research, Safety/ First Aid, operator's licenses)	4	34
Number of Indigenous persons trained in association with delivering the project (in any area of skills development/ expertise to enable the project to proceed including fisheries or habitat assessment, monitoring, research, Safety/ First Aid, operator's licenses)	1	1
For Commercial recipients only		
Percentage (%) of growth in annual sales		
Number of new employment (net) opportunities created through the project		

Commented [NCB1]: Not applicable because we don't have any full-time salaried employees. Everyone involved in the project is spending part of their time on it, or are students. Thus some of the lines that follow are 0 because we don't have anyone full time being paid from project funds

6. Project Activity-related Metrics [select all that apply]:

☒Project is a research or academic study resulting in published or publicly released studies/articles or data.

If selected, provide the number of publications produced/datasets released this year: 2

☒Project supports adoption/adaption of new or innovative products, processes, technologies, or equipment (i.e., a new process, technology, or delivery arrangement with important environmental implications). Note: a brief summary of the innovation used in, or stemmed from, your project is requested in the "Executive Summary" (Section 7 of this form).

If selected, provide the number of innovative products, processes, technologies, or equipment adopted/adapted this year: 2

☐Project supports protection and conservation of wild Pacific salmon

☐Project addresses modifications to the environment (e.g., water diversion, land use changes, habitat fragmentation or destruction, landslides, wildfires, etc.)

☐Project undertakes watershed assessment and planning toward ecosystem sustainability (e.g., habitat surveys, inventory or monitoring, indicator analysis)

☐Project results in fish habitat creation, enhancement or recovery (development of habitat prescriptions or engineering designs; physical restoration or rebuilding of habitat)

☐Project incorporates Indigenous Knowledge (IK)

☐Project supports stock enumeration or enhancement (salmon releases)

☒Project supports stock simulation modelling, scientific monitoring or data collection activities that inform fisheries management approaches, systems or frameworks for assessment and management of fisheries resources

☐Project improves air or water quality (e.g., through the reduction of chemical usage, improvements in fuel efficiency, greening of infrastructure)

☐Project supports commercial aquaculture

☐Project supports hatchery infrastructure improvements

☒Project contributes to the improved sustainability of fishing practices

☐Project supports fish health analysis (e.g., genetic analysis, virology, pathogen assessment)

☐Project improves understanding of ecosystem shifts and their impacts and/ or addresses impacts of climate change through assessment and monitoring

☐Project addresses threats from aquatic invasive species (AIS)

☐Project leads to improved product traceability, Eco-Certification or other internationally accepted standard to support increased quality or access to markets

Section 4: Fish and Fish Habitat Performance Metrics

If your project targets benefits to an identified Species At Risk (SAR) and/or contributes to fish habitat restoration or rebuilding in BC, please answer the survey questions in this section. If not, please proceed to Section 5.

The purpose of this section is to garner basic data about projects that are contributing to the sustainability of wild Pacific salmon and other BC fish stocks, specifically those have linkages to species at risk (SAR) in British Columbia and projects that address fish habitat decline. Data related to the project metrics this section are collected to enable BCSRIF to report out on how restoration project funding is contributing to rebuilding fish habitat in BC, consistent with annual recipient reporting requirements for other regional Grants and Contributions programs. Where exact numbers are unavailable, we ask that you estimate answers to the best of your knowledge.

For ease of reference, internet links to the federal and provincial SAR are provided below:

https://www.canada.ca/en/environment-climate-change/services/species-risk-public-registry.html

http://a100.gov.bc.ca/pub/eswp/

7. Identify the Species at Risk populations specifically targeted by your project [select all that apply]:

☐Southern BC chinook

☐Interior-Fraser coho

☐Fraser sockeye

☐Thompson-Chilcotin steelhead

☒Other SAR listed by the Committee on the Status of Endangered Wildlife in Canada (COSEWIC)

Specify: Inside and outside Yelloweye Rockfish; Quillback Rockfish, Bocaccio

☐Other species that is red or blue-listed by the BC Conservation Data Center (CDC)

Specify:____

8. Habitat impact/ limiting factor being addressed by project [select all that apply]:

☐Riverine or stream habitat loss/ decreased quality (e.g., degraded, damaged or destroyed channel structure, complexity (e.g., LWD) or functionality (e.g., pool: riffle structure)

☐Estuarine or nearshore marine habitat loss/ decreased quality (e.g., degraded salt marsh, intertidal/ foreshore areas)

☐Floodplain habitat connectivity loss (e.g., loss of fish access to seasonal wetlands, off-channel habitat, or side-channels)

☐Fish passage loss (e.g., barrier to fish migration, including seasonal or periodic blockages)

☐Riparian area loss or impacts (e.g., degraded, damaged or destroyed)

☐Aquatic invasive species (AIS)

☐Predation/ competition/ disease

☐Stream flow (inadequate flow or other adverse hydrology changes from for example, water extraction, flood-drought/ climate change impacts)

☐Stream substrate (erosion or sedimentation and associated impacts)

9. Fish Habitat Performance Measures

Performance Measures	Total Number for the Reported Fiscal Year	Cumulative Total *
Number of watershed or area recovery plans established (i.e. coastal, regional, sub-regional or watershed plans)		
Number of fish habitat restoration sub-projects *completed* (i.e. separate project sites). Note these should be listed in Section 1 (Project Information).		
Number of separate sites with engineered design drawings paid for, or partially paid for, with BCSRIF funding		
Fish Habitat-related Metrics		
Total area of fish habitat physically restored	(m2)	(m2)
- Total *aquatic habitat* (below high water mark, including marsh and wetland)	(m2)	(m2)
- Total *riparian habitat* (above high water mark, including foreshore and upland)	(m2)	(m2)

** if measurements are not in m2, please identify unit of measure*

10. Land Use(s) associated with the project restoration area(s) [select all that apply]:

☐Agriculture

☐Forestry

☐Industry (e.g., mining, oil/ gas, utilities, water-sewage facilities)

☐Urban

☐Transportation

☒Recreational

Section 5: Communication Activities

Information on your communication activities and the other standardized metrics in this section will be used to support intended program outreach targets and outcomes. This information provides valuable

insight into the scope and reach of project activities, and may be used by the Department to increase awareness of the BCSRIF program, as well as results achieved by recipients.

11. Communication Metrics

Communications Activities	Total Number for the Reported Fiscal Year	Cumulative Total
Number of Management Committee meetings held (if a Project Management Committee has been established as per the Articles of Agreement in your Contribution Agreement)	12	39
Number of workshops or other education or outreach events held with First Nations, stakeholders, or other public for engagement on project	8	32
Number of persons attending workshops or outreach events	70 (estimate)	1050
Number of training sessions delivered in association with the project (excluding outreach events, above)	0	1
Number of public communication products developed (e.g. reports, websites, videos, newsletters, brochures, information packages, training materials, social media posts)	20	128

12. Communication activity reference information

Please provide citation or relevant web link for the communication activity in the following table, as applicable to your project.

Communication Type (e.g. peer reviewed publication, reports, web publication, newsletter, magazine article, blog or media/ public event)	Communication Details (e.g. document title, headline, label, advertising)	Date of publication or release	Document citation or web link
Rockfish Conservation Project Facebook	Various social media posts related to rockfish	Ongoing	https://www.instagram.com/galianoconservancy/ https://www.facebook.com/galianoconservancy https://www.facebook.com/rockfishconservation

page (led by GCA)			
In person outreach	Galiano Saturday market	May – Sept 2022	Sharing info about rockfish at the Galiano Conservancy Association's booth at the weekly Saturday market on Galiano Island
Interview	CBC Radio (French)	October 2022	https://ici.radio-canada.ca/nouvelle/1922495/oceans-peches-sebaste-ile-galiano-environnement
Email announcements	Angler's Atlas listserv (~50000 subscribers), providing project updates	June 1, 22, 29, 20022	NA
Print magazine	Rockfish Conservation Area Maps in magazine	June 2022	NA (print copy available upon request)
Print magazine	Rockfish research promo in magazine	June 2022	NA (print copy available upon request)
Print magazine	Rockfish ID in magazine	June 2022	NA (print copy available upon request)
Conference presentation (152nd annual meeting of the American Fisheries Society)	Angler compliance with in Rockfish Conservation Areas in British Columbia, Canada, in response to awareness efforts and COVID-19	August 2022	Franklin W, PA Venturelli, Z McDonald, D Haggarty, N Ban, F Juanes, C Miltner, S Simmons (2022) Angler compliance with in Rockfish Conservation Areas in British Columbia, Canada, in response to awareness efforts and COVID-19. American Fisheries Society, Spokane WA
Conference presentation (10th World Recreational Fisheries Conference)	Angler compliance with in Rockfish Conservation Areas in British Columbia, Canada, in response to awareness efforts and COVID-19	February 2023	Venturelli PA, Z McDonald, D Haggarty, N Ban, F Juanes, C Miltner, S Simmons (2023) Angler compliance with in Rockfish Conservation Areas in British Columbia, Canada, in response to awareness efforts and COVID-19. Tenth World Recreational Fisheries Conference, Melbourne, Australia
Journal publication	ICES Journal of Marine Science	October 2022	Ban, N. C., C. Miltner, C. Matthews, M. Ankenman, S. Stelte, D. Haggarty, H. L. Davies, P. A. Venturelli, and F. Juanes. 2022. Decrease in recreational fisher compliance during the COVID-19 pandemic: the case of Rockfish Conservation Areas. ICES Journal of Marine Science 79:2277-2285.
Journal publication	Fisheries	August 2022	Davies, H. L., S. Gross, D. R. Haggarty, and F. Juanes. 2022. Conserving Rockfishes: Barotrauma and Descending Devices in the Northeast Pacific. Fisheries 47:406-411.
[add rows, as needed]			

Section 6: Executive Summary - Project achievements, Products, Outcomes & Benefits.

Information in this section is intended to provide an Executive Summary of your projects achievements and capture the overarching outcomes of your project *TO DATE*. It is meant to highlight the key advancements made by your organization toward achieving results that align with BCSRIF's three program pillars and objectives: **innovation, infrastructure improvements** and **science partnerships**.

This section is meant to capture your project's main highlights, headlines, and sound-bytes, as well as knowledge and information on the tangible benefits associated with your project. You are requested to demonstrate how your project meets with the programs objectives through a short narrative on each topic (~ 150 words), as applicable to your project. This information will be used by BCSRIF to report on the excellent work being undertaken throughout British Columbia in support of the economic and environmental sustainability of the fish and seafood sector, and the conservation and restoration of wild Pacific salmon.

Please note: You may also be submitting a separate year-end annual report as an identified deliverable of your CA which reports specifically on the project's fiscal-year activities. This section is intended to capture the overall progress and results achieved from your project to date.

Executive Summary – Summary of the project highlights in brief:

In this section, please provide a brief narrative on the project achievements ***TO DATE*** (i.e. *not* forthcoming). This information should be a narrative of your project highlights to complement the information, metrics and measures that you have provided in the preceding sections.

13. Executive Summary at Year-End/ Project highlights (i.e. major project milestones and current status)

Since April 2020, we have held monthly project team video conferences; continued developing the functionality of the MyCatch app; carried out field work (after ongoing COVID delays) to work on the descending device and sound components of the project; completed annual reports for remote camera monitoring and interviews on Galiano Island, and had students continue with data analysis. We have identified and obtained viable data, and conducted analyses to assess angler compliance with spatial regulations in both time and space with respect to education/outreach efforts and the COVID-19 pandemic on Galiano Island. We published our findings in an academic article that summarizes the pre- and during-pandemic changes in compliance with Rockfish Conservation Areas around Galiano Island. This was published in August 2022 in the ICES Journal of Marine Science, a prestigious international journal. We also published a second journal article, in the journal Fisheries, showcasing the research on barotrauma as a photo journal article.

Two notable innovations have emerged from this project. The first is an automated notification system alerting anglers as they move into a Rockfish Conservation Area when they are using the MyCatch app. When an angler passes into an RCA, an automatic notification appears on the top of map screen alerting the angler to this event. Additional information is then provided explaining the importance of using a descending device and how to ID Rockfish species. Ongoing updates continue to improve this functionality. Another innovation is the real time updates to fisheries regulations provided on the app. Although this is not the first app to do this, this is groundbreaking compared to many jurisdictions. The major innovation developed during this fiscal period was the automated notification system that will alert anglers when they approach an RCA. Ongoing updates to the app are helping to keep the information up-to-date.

Project Benefits – Summary of the project benefits in brief:

In this section, please provide a brief narrative on the project benefits ***TO DATE*** (i.e. *not* forthcoming). This information should complement the information, metrics and measures that you have provided in the preceding sections as a narrative of your project related to the ecological, socio-economic and partner/ community benefits realized to date, if applicable.

14. Ecological benefits realized to date

Our development of the MyCatch App, and ongoing outreach and education efforts BC-wide (e.g., through magazines, social media, listserv postings) and in-person on Galiano Island, are aimed at improving compliance with recreational guidelines, especially Rockfish Conservation Area.

15. Socio-Economic benefits realized to date

Anglers have been asking for better access to information regarding the recreational fishing regulations, especially boundaries of spatial closures such as Rockfish Conservation Areas. We are now able to provide this information through the MyCatch App. Continued in-person outreach on Galiano island and beyond, while more limited as the COVID-19 pandemic continues to restrict in-person events, continues to reach and educate coastal residents and anglers.

16. Benefits to partners and communities realized to date

Added capacity through this project has allowed Angler's Atlas to enhance the MyCatch App with functionality that has been requested by anglers. This has allowed Angler's Atlas to expand its reach into BC's coastal waters. The partnership with the Galiano Conservancy Association allows the continuation of the long-standing partnership in outreach and education, and research, between GCA and the University of Victoria, through hiring summer field technicians. The universities – University of Victoria and Ball State University – are benefiting from the partnerships to support students, and to expand reach to beyond the universities.

Project Innovations – Summary of the project highlights in brief

In this section, please provide a brief narrative on the project innovations ***TO DATE*** (i.e. *not* forthcoming). This information should complement the information, metrics and measures that you have provided in the preceding sections as a narrative of your project innovation(s), if applicable.

17. Innovation in implementing or resulting from your project to date

Two notable innovations have emerged from this project. The first is an automated notification system alerting anglers as they move into a Rockfish Conservation Area when they are using the MyCatch app. When an angler passes into an RCA, an automatic notification appears on the top of map screen alerting the angler to this event. Additional information is then provided explaining the importance of using a descending device and how to ID Rockfish species. Ongoing updates continue to improve this functionality. Another innovation is the real time updates to fisheries regulations provided on the app. Although this is not the first app to do this, this is groundbreaking compared to many jurisdictions. The major innovation developed during this fiscal period was the automated notification system that will alert anglers when they approach an RCA. Ongoing updates to the app are helping to keep the information up-to-date.

18. Please identify any additional Supporting Documentation you are submitting in association with the year-end report form in the box below (*if applicable*):

Annex #	Document Title	Attachment (electronic file name)
[add info]		
[add info]		
[add rows, as needed]		

Section 7: Narratives on Impacts to Projects and Program Improvements

The next two questions are meant to capture information regarding any challenges your organization faced in delivering your project over the past year (e.g. COVID-related impacts; natural-related disasters,

economic implications, securing support from other partners, etc.), and the measures undertaken to address those issues. The comments box is intended to allow recipients to provide feedback regarding the administration of the BCSRIF program and to suggest improvements to this program and/or future Grants and Contributions programs.

19. Challenges and lessons learned

Several aspects of our project have continued to be affected by COVID-19: in-person events continue to be limited, and logistics to do field work continue to be more complicated. We are also continuing to have challenges with recruitment and retention of students at Ball State University. We brought on a new student join us in 2022/23, but the student exited their studies. Ball State has thus continue to operating without a graduate student (undergraduate assistance only) since March 2021. We are progressing the work through other collaborations, as insufficient time remains for bringing on another student.

20. Other comments or information, and/or suggestions for improvement to the BCSRIF program

[Insert text in textbox]

Section 8: Confirmation

BCSRIF Project Number:	BCSRIF_2029_006
Project Title:	Enhancing rockfish recovery through citizen science, outreach & field experiments
Recipient:	University of Victoria
Name and title of authorized official:	Natalie Ban, Professor
Date:	26 April 2023